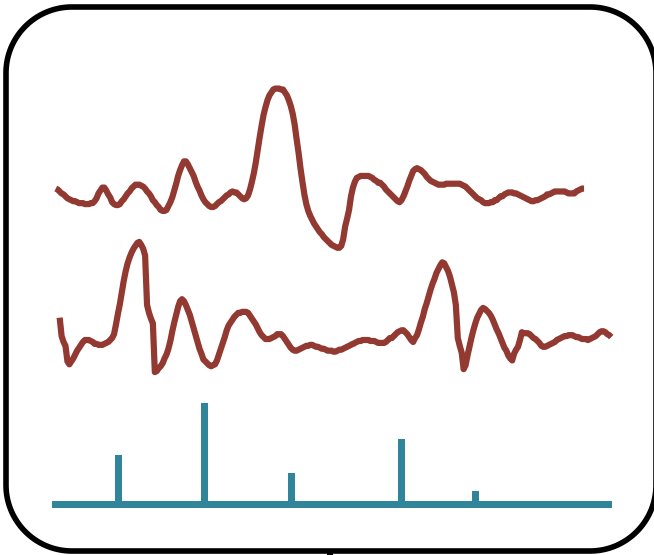


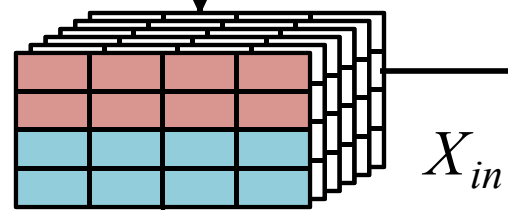
DDM Readings



Preprocessing

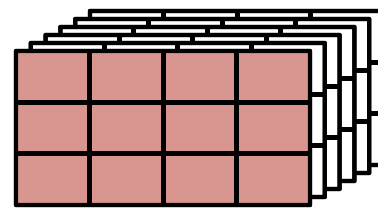
Feature Construction

Input Views



Dual-Path Feature Embedding

Sensor Features



x_t

Sensor Measurement Branch

Sensor ID

Quantile encoding

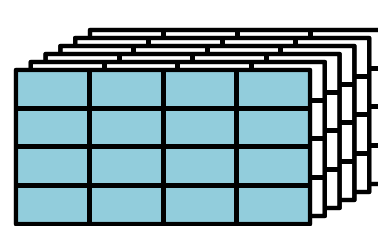
Linear projection

Quantiles

Type ID

Sensor tokens

Statistical Features



s_t

Statistical Feature Branch

Group IDs + Type ID

Normalize

asinh

Level / range

Variability

Relations

Linear

Linear

Linear

Statistical tokens

Training samples

Sensor Interaction Transformer

Joint tokens

LayerNorm

Multi-Head Attention

LayerNorm

Feed Forward

Updated tokens

Learnable module token
Statistical group tokens

Risk Head

Inference

Module readout

Linear

Sigmoid

Failure Probability

Hierarchical Sample Selection

Training Only

Normal windows

High risk

Coverage

Random

Selected windows